Christina Willecke Lindberg

J.C. Ryan Prize Postdoctoral Fellow

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RESEARCH INTERESTS

Local galaxies ○ Interstellar Medium ○ Stellar populations ○ HST ○ JWST ○ Roman ○ Data science ○ Community leadership

APPOINTMENTS

Inaugural J.C. Ryan Prize Postdoctoral Fellow 2025 - Present
Center for Astrophysics | Harvard & Smithsonian (CfA)

EDUCATION

Ph.D. Candidate in Astronomy 2019 - 2025
Johns Hopkins University (JHU)
Thesis Topic: Resolving the Multi-Scale Structure of the Interstellar Medium in Local Group Galaxies
Research Advisor: Dr. Claire Murray (STScI)

B.S. in Comprehensive Physics and Astronomy (Honors) 2015 - 2018
University of Washington (UW)

SOFTWARE

Bayesian Extinction And Stellar Tool (BEAST) 2019 - Present
Active Developer
Open-source Python package for probabilistically modeling dust-extinguished multi-band spectral energy distributions of resolved stellar populations in nearby galaxies.

AsteroGaP 2018 - 2021
Lead Developer
Open-source Bayesian Python package for modeling sparse asteroid light curve profiles with Gaussian Processes and Markov Chain Monte Carlo models for Zwicky Transient Facility and Rubin Observatory data.

OBSERVING PROGRAMS AS PI

Hubble Space Telescope Cycle 31 - 15 orbits
2023 PI: Winging the SMC: 3D Structure of the Interstellar Medium in the Tidally Disrupted Wing of the SMC

James Webb Space Telescope Cycle 4 - 12 hours orbits
2023 PI: Winging the SMC: 3D Structure of the Interstellar Medium in the Tidally Disrupted Wing of the SMC

OBSERVING PROGRAMS AS CO-I

Roman Space Telescope Cycle 1
2026 Co-I: A Legacy Survey of Andromeda and Triangulum with Roman - GAS

Roman Space Telescope Cycle 1
2026 Co-I: Foundational Value-Added Data Products for the Roman Galactic Plane Survey - Analysis

Hubble Space Telescope Cycle 33 - Archival
2025 Co-I: Unraveling the Tarantula's Web: A precise, high-resolution dust map of 30 Doradus

James Webb Space Telescope Cycle 3 - 25.8 hours
2024 Co-I: Zooming-in on the sub-grid physics of PAHs at 20% solar metallicity

Hubble Space Telescope Cycle 32 - 162 orbits
2024 Co-I: Bringing HST to the VLA: The Interaction of Stars and Gas in the Local Group

Hubble Space Telescope Cycle 30 - 12 orbits
2022 Co-I: Taming the BEAST of N66 to resolve how star formation shapes the interstellar medium at low metallicity

Hubble Space Telescope Cycle 26 - 73 orbits
2018 Co-I: QuaStar: The first unobscured view of the Milky Way's Circumgalactic Medium

FIRST-AUTHOR PUBLICATIONS

5. **C. W. Lindberg**, et al., "Scylla VII: Observational Evidence for an Order of Magnitude in Dust Opacity Evolution with ISM Density in the Large Magellanic Cloud", 2026, *in prep for ApJ Letters*
4. **C. W. Lindberg**, et al., "Scylla VI: Parsec-Scale Extinction Maps in the Magellanic Clouds", 2026, *accepted to ApJ*
3. **C. W. Lindberg**, et al., "Scylla IV: Intrinsic Stellar Properties and Line-of-Sight Dust Extinction Measurements Towards 1.5 Million Stars in the SMC and LMC", 2025, *ApJ, 982, 33*
2. **C. W. Lindberg**, C. Murray, J. Dalcanton, J. Peek, K. Gordon, "Dust around massive stars is agnostic to galactic environment", 2024, *ApJ, 963, 58*
1. **C. W. Lindberg**, D. Huppenkothen, R. L. Jones, B. T. Bolin, M. Jurić et al., "Characterizing Sparse Asteroid Light Curves with Gaussian Processes", 2022, *AJ, 163, 29*

OTHER PUBLICATIONS (†= significant contribution)

10. I. Escala et al. incl. **C. W. Lindberg**, "Scylla at APOGEE: The Impact of Starbursts on the Chemical Evolution of the Magellanic Clouds", 2026, *accepted by ApJ*
9. E. Tarantino et al. incl. **C. W. Lindberg**, "JWST Captures Growth of Aromatic Hydrocarbon Dust Particles in the Extremely Metal-poor Galaxy Sextans A", 2025, *submitted to Nature*
8. C. Burhenne et al. incl. **C. W. Lindberg**, "Scylla V: Constraints on the spatial and temporal distribution of bursts and the interaction history of the Magellanic Clouds from their resolved stellar populations", 2025, *accepted to ApJ*
7. K. Gilbert et al. incl. **C. W. Lindberg**, "The Local Ultraviolet to Infrared Treasury I. Survey Overview of the Broadband Imaging", 2024, *ApJS, 276, 8*
6. R. Cohen et al. incl. **C. W. Lindberg**, "Scylla III. The Outside-In Radial Age Gradient in the Small Magellanic Cloud and the Star Formation Histories of the Main Body, Wing and Outer Regions", 2024, *ApJ, 975, 43*
5. R. Cohen et al. incl. **C. W. Lindberg**, "Scylla II. The Spatially Resolved Star Formation History of the Large Magellanic Cloud Reveals an Inverted Radial Age Gradient", 2024, *ApJ, 975, 42*
4. †C. Murray, **C. W. Lindberg**, et al., "Scylla I: A pure-parallel, multi-wavelength imaging survey of the ULLYSES fields in the LMC and SMC", 2024, *ApJS, 275, 5*
3. T. Wainer et al. incl. **C. W. Lindberg**, "PHATTER. VI. The High-Mass Stellar IMF in M33", 2024, *ApJ, 168, 86*
2. C. Murray et al. incl. **C. W. Lindberg**, "A Galactic Eclipse: The Small Magellanic Cloud is Forming Stars in Two, Superimposed Systems", 2023, *ApJ, 962, 120*
1. B. Williams et al. incl. **C. W. Lindberg**, "The Panchromatic Hubble Andromeda Treasury: Triangulum Extended Region (PHATTER) I. Ultraviolet to Infrared Photometry of 22 Million Stars in M33", 2021, *ApJS, 253, 53*

ACADEMIC SERVICE

Time Allocation Committee Member, *Submillimeter Array* 2026 - Present
Referee, *The Astrophysical Journal* 2025 - Present
Review panelist, *Fornax External System Readiness Review Panel* 2025
Local Organizing Committee Member, *STScI Spring Symposium - INTER+STELLAR* 2025

LEADERSHIP EXPERIENCE

JHU Physics and Astronomy Graduate Students (PAGS) - President 2022 - 2023

Represented +130 graduate students and coordinated initiatives to improve the graduate experience within the JHU Physics and Astronomy Department e.g. pay raises, travel grants, professional society memberships, mentorship program, etc. Led discussions with the department chair and graduate program committee to improve faculty-student communication and gender equity.

CfA Research Academy (CARA) 2025 - Present

Organized graduate school application workshops for postbaccalaureate and master's students, and served as panelist for discussion on navigating the academic path — research, mentoring, publishing, funding, work–life balance, etc.

Astronomy Graduate Congress - JHU Representative 2024 - 2025

Served as JHU representative for the Astronomy Graduate Congress, which provides a common platform for graduate students to discuss issues regarding graduate education in astronomy.

Gender Minorities and Women in Physics (GWIP) JHU Chapter - LOC Member 2023 - 2025

Helped organize +100 person annual GWIP Summit aimed at fostering connections among women and gender minorities in physics across all career stages and institutions in the DMV region.

MENTORING

Research Advising

Akmal Hashad - Harvard Undergraduate (Spring 2026 - Present)

JHU PHA Mentorship Program

Sasha Levina - First-year graduate student (Fall 2023 - Spring 2024)
Qiushi (Chris) Tian - Post-bac student (Fall 2024 - Spring 2025)
Alexia Knight - First-year undergraduate student (Fall 2024 - Spring 2025)

TEACHING

Johns Hopkins University

AS 173.111/112 *General Physics Lab I & II* 2019-2020
AS.171.101/108 *General Physics: Physical Science/Electromagnetism (Active Learning)* 2019-2020

University of Washington

ASTR 101/150 *Introductory Astronomy/The Planets* 2018 - 2019
PHYS 122 *Electromagnetism* 2018

GRANTS & AWARDS

APS Women in Physics Group Grant (Co-author: $1,000) 2025

James Webb Space Telescope Cycle 4 (PI: $220,709) 2024

Hubble Space Telescope Cycle 31 (PI: $134,040) 2024

AAS FAMOUS Travel Grant ($1,000) 2022

UW Mary Gates Research Scholarship 2018

SELECTED PRESENTATIONS

KIPAC Seminar 2025
Talk: Mapping the ISM at Parsec-Scales with Dust Extinction in the SMC and LMC

STScI Spring Symposium: INTER+STELLAR 2025
Talk: Mapping the ISM at Parsec-Scales with Dust Extinction in the SMC and LMC

Institute for Advanced Science Astrocoffee 2024
Talk: Dust around massive stars is agnostic to galactic environment: New insights from PHAT & BEAST

Princeton Thunch 2024
Seminar: Reading Between the Stars: Resolving the Multi-Scale Interstellar Medium in Local Group Galaxies

UC San Diego A&A Journal Club 2024
Talk: Scylla IV: Intrinsic Stellar Properties and Line-of-Sight Dust Extinction Measurements Towards 1.5 Million Stars in the SMC and LMC

Harvard-Smithsonian CfA ITC Luncheon 2024
Seminar (Invited): Reading Between the Stars: Resolving the Multi-Scale Interstellar Medium in Local Group Galaxies

Columbia Astronomy 2024
Seminar: Reading Between the Stars: Resolving the Multi-Scale Interstellar Medium in Local Group Galaxies

Rutgers Astrocoffee 2024
Talk: Dust around massive stars is agnostic to galactic environment: New insights from PHAT & BEAST

Astronomy on Tap - Baltimore 2024
Talk: Massive Stars and Where to Find Them

HotSci@JHU/STScI 2024
Talk: The Future of HST - Tracing the Multi-Phase ISM via Dust Extinction in Nearby Galaxies with UV Photometry

STScI Spring Symposium: Recipes to Regulate Star Formation at All Scales 2024
Poster: Dust around massive stars is agnostic to galactic environment: New insights from PHAT & BEAST

Flatiron Institute CCA: XMC Workshop 2024
Talk: Constraining the 3D Structure of the ISM in 30 Doradus with Scylla

Resolving Galaxy Ecosystems Across All Scales 2023
Plenary Talk: Dust Around Massive Stars is Agnostic to Galactic Environment

237rd Meeting of the American Astronomical Society 2021
Poster & Talk: Investigating Massive Stars in M31

236rd Meeting of the American Astronomical Society 2020
Poster: Studying Nearby Low-Metallicity Environments with Scylla

233rd Meeting of the American Astronomical Society 2019
Talk: A Bayesian-Based Method for Inferring Asteroid Properties from Sparse Light Curve

50th Meeting of the Division of Planetary Sciences 2018
Talk: A Bayesian-Based Method for Inferring Asteroid Properties from Sparse Light Curve

Mary Gates 21st Annual Undergraduate Research Symposium 2018
Talk: Werk SQuAD: The Quest to Better Understand Galaxies and Their Surrounding Medium

231rd Meeting of the American Astronomical Society 2018
Poster: Classifying Variable Sources in SDSS Stripe 82